MATH 128a: Homework #7

Instructor: F. Weber

Due by: **Wednesday, October 22, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 2.5

Solve problems 1d, 4, 6, 9, 15, 17 of Section 2.5 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 2: Newton's method

The function

$$f(x) = \frac{x}{\sqrt{1+x^2}}$$

has a unique root at $x = 0$.

a) Show that Newton's method gives $x_{k+1} = -x_k^3$ (where x_k is the approximation to the root obtained in the kth iteration and x_{k+1} is the approximation obtained in the $k+1$th iteration.). Explain why the method succeeds in approximating the root if and only if $|x_0| < 1$.

b) Draw graphs to illustrate the first few iterates when $x_0 = 0.5$ and when $x_0 = 1.5$.

Problem 3: Stopping criterion for fixed point iteration

Let $\Phi(x) = x$ a fixed point equation with a fixed point x^* which is not explicitly known, and a sequence x_k, $k \geq 1$ computed using the nonlinear fixed point iteration from some starting value x_0. Assume that the sequence x_k converges to x^* as $k \to \infty$ and that $|\Phi'(x)| \leq q < 1$ for all $x \in \mathbb{R}$. If you would like to get an approximation x_K for some $K \in \mathbb{N}$ to the root x^* that satisfies $|x^* - x^K| < \tau$ for some given tolerance $\tau > 0$, what is a good stopping criterion for the iteration? Explain your choice with rigorous arguments. The expression should NOT depend on x^*.